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(57) Abstract:

ABSTRACT MULTI-AGENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND AUTOMATED INTERVENTION A student performance analysis system (100) comprises a data processing module (110) configured to receive structured academic data associated with a plurality of students and generate enriched student profile data, and a dual-tier memory architecture (120) comprising a learner memory unit (122) having an episodic memory component (122a), a semantic memory component (122b), and a working memory component (122c), and a class memory unit (124). The system (100) comprises specialized agents (130) including a goal agent (132), diagnosis agent (134), path planner agent (136), feedback agent (138), and analytics agent (140), each writing outputs to the working memory component (122c). An educational orchestrator (150) executes the specialized agents (130) in a predetermined pedagogical sequence. An agent communication layer (160) mediates inter-agent communication and maintains audit logs. An output generation module (170) generates personalized reports based on outputs in the working memory component (122c).

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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**&
THE PATENTS RULES, 2003**

**COMPLETE
SPECIFICATION
(See section 10 and rule 13)**

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TITLE OF INVENTION

**MULTI-AGENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS
AND AUTOMATED INTERVENTION**

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**25 THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to educational analytics systems employing artificial intelligence and machine learning, and more particularly to a
10 multi-agent system comprising specialized agents operating over a dual-tier memory architecture for student performance analysis and automated academic intervention.

BACKGROUND

[0004] The following text of background art is provided for purposes of
15 understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Educational institutions increasingly rely on data-driven approaches to monitor and enhance student learning outcomes. As class sizes grow and educational environments become more complex, the ability to provide timely,
20 personalized academic feedback to each student presents a persistent challenge. Traditional methods of performance assessment often fail to scale effectively, leaving instructors unable to deliver individualized guidance to every learner.

[0006] Various systems have been developed to address aspects of this challenge. US11380211B2 discloses a personalized mastery learning platform that
25 adapts educational activities based on learner progress. US10490092B describes an intelligent system employing data-driven learning techniques. US12165655B1 presents a conversational AI system that engages in Socratic-style questioning for teaching, testing, or evaluating participants. US12272265B2 discloses an adaptive and scalable AI-driven personalized learning platform utilizing AI and graph
30 database technology to contextualize educational content for learners using large language models. US11443648B2 describes systems and methods to assist course

instructors through adaptive learning that uses learner behavior analytics and machine learning to select and personalize learning content.

5 [0007] Additional approaches have focused on predictive analytics and content delivery. US12100315B2 discloses a peer-inspired student performance prediction system using graph neural networks to predict student performance in unattempted questions within interactive online question pools. US11620918B2 describes delivering personalized learning material based on a student's comprehension level. US10467919B2 presents an expert system software that helps students answer questions and assists instructors in creating assessments. Reddy et al. (2026)
10 describes an AI learning assistant that provides assistance but lacks an integrated analytics and notification system.

[0008] While these approaches address certain aspects of educational analytics, there remains a gap in providing comprehensive systems that combine personalized intervention with structured reasoning across multiple analytical tasks.
15 Existing solutions may conflate separate reasoning tasks such as goal-setting, diagnosis, planning, and communication, and may lack structured intermediary artifacts that can be inspected, audited, or used by subsequent processing components. Furthermore, there is a desire for systems capable of shared reasoning across students with diagnosed conceptual gaps while prescribing personalized
20 action plans.

[0009] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

OBJECT

25 [0010] In view of the foregoing, and without prejudice to the generality of the disclosure, it is an object of the present invention to provide. It is to be understood that the foregoing objects are illustrative and non-limiting; additional objects will be apparent from the description, examples, and claims, and need not be achieved in every embodiment.

30 [0011] The object of the present disclosure is to provide a locally deployable multi-agent-based reasoning system that generates personalized feedback for

students based on their academic performance. The system aims to analyze student performance using context-aware reasoning enabled by large language models operating over a structured memory architecture.

5 [0012] A further object of the present disclosure is to provide a system capable of identifying learning gaps and generating personalized recommendations while maintaining memory isolation between individual students. The system seeks to enable automated report generation and feedback dispatch to each student without manual intervention.

10 [0013] Yet another object of the present disclosure is to provide class-level insights for instructors, enabling educators to identify at-risk students, monitor performance trends, and prioritize interventions across a student cohort. The system aims to create a scalable and intelligent academic intervention framework that sequences specialized agents in a pedagogical order over a dual memory layer, with an audit-logged communication mechanism that ensures every analytical decision
15 is traceable and explainable.

[0014] For the avoidance of doubt, the foregoing objects encompass compositions and manufacturing processes suitable for adoption in jurisdictions where claims to methods of treatment of animals may be restricted; corresponding “use” or “product” claim formats are contemplated without limitation.

20 SUMMARY

[0015] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and
25 accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0016] According to an aspect of the present disclosure, a student performance analysis system (100) is provided. The system comprises a data processing module
30 (110) configured to receive structured academic data associated with a plurality of students and generate enriched student profile data. The system further comprises

a dual-tier memory architecture (120) comprising a learner memory unit (122) associated with each student of the plurality of students. The learner memory unit (122) comprises an episodic memory component (122a) configured to store historical academic events, a semantic memory component (122b) configured to store derived performance metrics, and a working memory component (122c) configured to store intermediate and final analysis outputs. The dual-tier memory architecture (120) further comprises a class memory unit (124) configured to store aggregated class-level data. The system comprises a plurality of specialized agents (130) comprising a goal agent (132), a diagnosis agent (134), a path planner agent (136), a feedback agent (138), and an analytics agent (140), wherein each specialized agent is configured to perform a distinct analytical function and write outputs to the working memory component (122c). The system further comprises an educational orchestrator (150) configured to execute the plurality of specialized agents (130) in a predetermined pedagogical sequence. The system comprises an agent communication layer (160) configured to mediate inter-agent communication and maintain audit logs of agent interactions. The system also comprises an output generation module (170) configured to generate personalized reports based on the outputs stored in the working memory component (122c).

[0017] The dual-tier memory architecture separating learner-specific memory from class-level memory enables memory isolation between individual students while permitting aggregated analysis across the student cohort. The use of specialized agents each performing a distinct analytical function allows for modular reasoning where each analytical task is handled independently, producing structured intermediary artifacts that can be inspected, audited, or used by subsequent processing components. The predetermined pedagogical sequence ensures that agents operate in a logical order that mirrors effective educational intervention workflows. The agent communication layer with audit logs provides traceability and explainability for every analytical decision made by the system.

[0018] According to other aspects of the present disclosure, the feedback agent (138) may comprise a large language model configured to generate personalized feedback messages based on data retrieved from the learner memory unit (122).

[0019] The use of a large language model within the feedback agent enables context-aware reasoning and natural language generation, allowing the system to produce personalized feedback messages that are tailored to each student's specific academic profile and learning history stored in the learner memory unit.

5 [0020] According to other aspects of the present disclosure, the agent communication layer (160) may be configured to provide typed message exchange between the plurality of specialized agents (130), queue-based communication handling, and timestamped communication records.

[0021] Typed message exchange ensures that communications between agents
 10 conform to defined data structures, reducing errors and enabling reliable inter-agent coordination. Queue-based communication handling allows for asynchronous processing and orderly message delivery. Timestamped communication records provide a complete audit trail that supports transparency and enables post-hoc analysis of the system's reasoning process.

15 [0022] According to other aspects of the present disclosure, the goal agent (132) may be configured to define measurable academic objectives, the diagnosis agent (134) may be configured to identify knowledge gaps and conceptual weaknesses, and the path planner agent (136) may be configured to generate structured improvement plans comprising topic prioritization and study sched

20 [0023] According to an aspect of the present disclosure, a student performance analysis system (100) is provided. The system comprises a data processing module (110) configured to receive structured academic data associated with a plurality of students and generate enriched student profile data. The system further comprises a dual-tier memory architecture (120) comprising a learner memory unit (122)

25 associated with each student of the plurality of students. The learner memory unit (122) comprises an episodic memory component (122a) configured to store historical academic events, a semantic memory component (122b) configured to store derived performance metrics, and a working memory component (122c) configured to store intermediate and final analysis outputs. The dual-tier memory

30 architecture (120) further comprises a class memory unit (124) configured to store aggregated class-level data. The system comprises a plurality of specialized agents

(130) comprising a goal agent (132), a diagnosis agent (134), a path planner agent (136), a feedback agent (138), and an analytics agent (140), wherein each specialized agent is configured to perform a distinct analytical function and write outputs to the working memory component (122c). The system comprises an educational orchestrator (150) configured to execute the plurality of specialized agents (130) in a predetermined pedagogical sequence. The system comprises an agent communication layer (160) configured to mediate inter-agent communication and maintain audit logs of agent interactions. The system comprises an output generation module (170) configured to generate personalized reports based on the outputs stored in the working memory component (122c).

[0024] The dual-tier memory architecture provides technical advantages by enabling memory isolation between individual students while maintaining aggregated class-level data, thereby allowing personalized analysis without cross-contamination of student records. The separation of episodic, semantic, and working memory components enables structured storage and retrieval of different types of academic information, facilitating efficient context-aware reasoning. The predetermined pedagogical sequence of specialized agents ensures that analytical tasks are performed in a logical order that mirrors educational reasoning processes, while the audit logs maintained by the agent communication layer provide traceability and explainability for every analytical decision made by the system.

[0025] According to other aspects of the present disclosure, the system (100) may include one or more of the following features. The feedback agent (138) may comprise a large language model configured to generate personalized feedback messages based on data retrieved from the learner memory unit (122).

[0026] The use of a large language model within the feedback agent enables the generation of natural language feedback that is contextually relevant to each student's individual academic history and performance metrics. This approach allows for nuanced and personalized communication that adapts to the specific learning context of each student, rather than relying on generic template-based responses.

[0027] According to other aspects of the present disclosure, the agent communication layer (160) may be configured to provide typed message exchange between the plurality of specialized agents (130), queue-based communication handling, and timestamped communication records.

5 [0028] The typed message exchange ensures that communications between agents conform to expected data structures, reducing errors and enabling reliable inter-agent coordination. Queue-based communication handling allows for asynchronous processing and orderly management of agent interactions, while timestamped communication records provide a chronological audit trail that
10 supports debugging, verification, and accountability of the analytical process.

[0029] According to other aspects of the present disclosure, the goal agent (132) may be configured to define measurable academic objectives. The diagnosis agent (134) may be configured to identify knowledge gaps and conceptual weaknesses. The path planner agent (136) may be configured to generate structured
15 improvement plans comprising topic prioritization and study schedules.

[0030] The separation of goal-setting, diagnosis, and planning functions into distinct specialized agents enables modular reasoning where each analytical task produces structured intermediary artifacts that can be inspected, audited, or used by subsequent processing components. This decomposition allows for targeted
20 improvements to individual reasoning components without affecting the overall system architecture, and ensures that each stage of the analytical pipeline produces verifiable outputs.

[0031] According to other aspects of the present disclosure, the analytics agent (140) may be configured to perform class-level analysis comprising performance
25 distribution analysis, identification of weak areas across the plurality of students, and generation of instructor-level recommendations.

[0032] The class-level analysis capability enables instructors to identify patterns and trends across an entire student cohort, facilitating the identification of at-risk students and common conceptual difficulties. This aggregated view supports
30 prioritization of interventions and resource allocation at the classroom level,

complementing the individualized student-level analysis performed by other agents.

5 [0033] According to other aspects of the present disclosure, the episodic memory component (122a) may be configured to store assessment attempts, scores obtained, submission time data, and difficulty-level interactions for each student.

[0034] The storage of detailed episodic data including assessment attempts, scores, submission timing, and difficulty-level interactions provides a comprehensive historical record of each student's academic journey. This granular data enables temporal analysis of learning patterns and supports the identification
10 of trends in student performance over time.

[0035] According to other aspects of the present disclosure, the semantic memory component (122b) may be configured to store a performance band classification, group-wise proficiency data, and a computed risk score for each student.

15 [0036] The semantic memory component enables efficient storage and retrieval of derived metrics that represent higher-level abstractions of student performance. The performance band classification, group-wise proficiency data, and computed risk scores provide readily accessible indicators that support rapid assessment and comparison without requiring recomputation from raw episodic
20 data.

[0037] According to other aspects of the present disclosure, the system (100) may further comprise a performance classification module (180) configured to classify each student into one of a plurality of performance bands based on an overall performance score.

25 [0038] The performance classification module enables systematic categorization of students based on their academic standing, facilitating targeted intervention strategies appropriate to each performance band. This classification supports scalable management of student cohorts by enabling batch processing and prioritization based on performance categories.

30 [0039] According to other aspects of the present disclosure, the output generation module (170) may be configured to generate at least one of a portable

document format report, a spreadsheet report, and an electronic mail notification comprising a student performance summary and a study roadmap.

5 [0040] The support for multiple output formats including portable document format reports, spreadsheet reports, and electronic mail notifications enables flexible delivery of personalized feedback through channels appropriate to different stakeholders and use cases. The inclusion of both performance summaries and study roadmaps in the generated outputs provides students with actionable guidance alongside their assessment results.

10 [0041] According to other aspects of the present disclosure, the educational orchestrator (150) may be configured to execute the goal agent (132), the diagnosis agent (134), the path planner agent (136), and the feedback agent (138) sequentially for each student, and to execute the analytics agent (140) after completion of individual student analyses.

15 [0042] The sequential execution of individual student-focused agents followed by the class-level analytics agent ensures that personalized analysis is completed before aggregation, enabling the analytics agent to operate on complete individual results. This execution order mirrors pedagogical reasoning processes where individual assessment precedes cohort-level analysis, and ensures that class-level insights incorporate the full set of individual student analyses.

20 [0043] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

25 [0044] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

30 [0045] The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that

the invention is not limited to the precise implementations shown in the figures and may encompass variations without departing from the scope of protection.

[0046] FIG. 1 illustrates a block diagram of a student performance analysis system, according to aspects of the present disclosure.

5 [0047] FIG. 2 depicts a dual-tier memory architecture of the system of FIG. 1, according to an embodiment.

[0048] FIG. 3 illustrates an interaction between specialized agents and a working memory component, according to aspects of the present disclosure.

[0049] FIG. 4 depicts an educational orchestrator and an agent communication
10 layer of the system of FIG. 1, according to an embodiment.

[0050] FIG. 5 illustrates a method for student performance analysis, according to aspects of the present disclosure.

[0051] FIG. 6 depicts the agent communication layer of the system of FIG. 1, according to an embodiment.

15 [0052] Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily apparent to those of ordinary skill in the art
20 having benefit of the description herein.

DETAILED DESCRIPTION

[0053] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to
25 the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0054] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and
30 should be interpreted to include, but not be limited to, the mentioned elements.

Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0055] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0056] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0057] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0058] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0059] FIG. 1 illustrates a block diagram of a student performance analysis system 100 according to various embodiments.

[0060] The student performance analysis system 100 comprises a data processing module 110 configured to receive structured academic data associated with a plurality of students and generate enriched student profile data. The data

processing module 110 may receive input from spreadsheet files containing student performance records. In some cases, the data processing module 110 processes data through a defined pipeline sequence comprising load_all_files, build_learner_memories, and build_class_memory functions. The data processing module 110 may be configured to perform extraction and normalization of lab identifiers and difficulty levels from the structured academic data. In some cases, the data processing module 110 may detect rapid or abnormal submissions based on submission duration data. The data processing module 110 may map labs into predefined groups for aggregation purposes.

10 **[0061]** The student performance analysis system 100 further comprises a dual-tier memory architecture 120. The dual-tier memory architecture 120 enables context-aware reasoning at both student and class levels. The dual-tier memory architecture 120 is described in further detail below.

15 **[0062]** The student performance analysis system 100 comprises a plurality of specialized agents 130. The specialized agents 130 comprise a goal agent, a diagnosis agent, a path planner agent, a feedback agent, and an analytics agent. Each specialized agent may be configured to perform a distinct analytical function. The specialized agents 130 are described in further detail below.

20 **[0063]** The student performance analysis system 100 comprises an educational orchestrator 150 configured to execute the plurality of specialized agents 130 in a predetermined pedagogical sequence. The educational orchestrator 150 may coordinate multi-agent processing and manage controlled access to memory structures. The educational orchestrator 150 is described in further detail below.

25 **[0064]** The student performance analysis system 100 comprises an agent communication layer 160 configured to mediate inter-agent communication and maintain audit logs of agent interactions. The agent communication layer 160 provides structured, traceable interaction between the specialized agents 130. The agent communication layer 160 is described in further detail below.

30 **[0065]** The student performance analysis system 100 comprises an output generation module 170 configured to generate personalized reports based on outputs stored in a working memory component of the dual-tier memory

architecture 120. The output generation module 170 may be configured to generate at least one of a portable document format report, a spreadsheet report, and an electronic mail notification comprising a student performance summary and a study roadmap.

- 5 **[0066]** In some cases, the output generation module 170 generates one-page personalized student portable document format reports containing charts, topic-wise scores, weak areas, agent outputs, and personalized feedback. The output generation module 170 may generate spreadsheet reports comprising full feedback sheets, lab group summaries, student-wise rankings, and class analytics sheets. In
- 10 some cases, the output generation module 170 generates band-specific personalized hypertext markup language email notifications containing a student performance summary, a study roadmap, motivational feedback, and call-to-action recommendations. The output generation module 170 may generate instructor dashboard insights comprising weak student identification, class trends,
- 15 intervention priorities, and lab performance bottlenecks.

- [0067]** The student performance analysis system 100 further comprises a performance classification module 180 configured to classify each student into one of a plurality of performance bands based on an overall performance score. The performance classification module 180 may classify students into four bands:
- 20 Excellent for scores greater than 90%, Good for scores between 80% and 90%, Average for scores between 70% and 80%, and Below Average for scores less than 70%. The performance classification module 180 may assign corresponding risk scores for prioritizing interventions based on performance band classification.

- [0068]** The student performance analysis system 100 may be configured for
- 25 local deployment without requiring cloud-based infrastructure. Local deployment enables the student performance analysis system 100 to operate on institutional computing resources without external network dependencies.

- [0069]** FIG. 2 illustrates the dual-tier memory architecture 120 according to various embodiments.

- 30 **[0070]** The dual-tier memory architecture 120 comprises a learner memory unit 122 and a class memory unit 124. The learner memory unit 122 may be

associated with each student of the plurality of students. The dual-tier memory architecture 120 enables context-aware reasoning at both student and class levels through separation of individual student data from aggregated class-level data.

[0071] The learner memory unit 122 comprises three logical components: an episodic memory component 122a, a semantic memory component 122b, and a working memory component 122c.

[0072] The episodic memory component 122a may be configured to store historical academic events. The episodic memory component 122a may store assessment attempts, scores obtained, submission time data, and difficulty-level interactions for each student. In some cases, the episodic memory component 122a stores behavioral patterns observed during student interactions with assessments. The behavioral patterns may include response timing characteristics, navigation sequences through assessment materials, and patterns of engagement with different difficulty levels. In some cases, the episodic memory component 122a may be implemented using a relational database structure with indexed fields for temporal queries. In other cases, the episodic memory component 122a may utilize a document-based storage format that preserves the sequential nature of academic events. The episodic memory component 122a may employ time-series data structures to facilitate analysis of performance trends over multiple assessment periods.

[0073] The semantic memory component 122b may be configured to store derived performance metrics. The semantic memory component 122b may store a performance band classification, group-wise proficiency data, and a computed risk score for each student. In some cases, the semantic memory component 122b stores difficulty-wise proficiency metrics for each student. The difficulty-wise proficiency metrics may indicate student performance across different levels of assessment complexity. The semantic memory component 122b may further store identification of strong and weak areas for each student. In some cases, the semantic memory component 122b may be implemented using key-value storage structures that enable rapid retrieval of computed metrics. In other cases, the semantic memory component 122b may utilize graph-based data structures that represent

relationships between proficiency areas. The semantic memory component 122b may employ caching mechanisms to reduce recomputation of derived metrics from raw episodic data.

5 **[0074]** The working memory component 122c may be configured to store intermediate and final analysis outputs. The working memory component 122c may store learning goals generated by the goal agent 132, diagnostic insights generated by the diagnosis agent 134, improvement plans generated by the path planner agent 136, and personalized feedback generated by the feedback agent 138. In some cases, the working memory component 122c may be implemented as a transient storage
10 structure that is populated during agent execution and cleared after report generation. In other cases, the working memory component 122c may persist analysis outputs for historical comparison across multiple analysis cycles. The working memory component 122c may utilize structured data formats that enable downstream processing by the output generation module 170.

15 **[0075]** The class memory unit 124 may be configured to store aggregated class-level data. The class memory unit 124 may store total number of students, distribution across performance bands, and group-level averages. In some cases, the class memory unit 124 stores detection of abnormal submission patterns across the plurality of students. The detection of abnormal submission patterns may
20 include identification of submissions with unusually short completion times or submissions that deviate from expected temporal patterns. In some cases, the class memory unit 124 stores identification of at-risk students based on aggregated performance data. The identification of at-risk students may be based on computed risk scores, performance band classifications, and trend analysis across multiple
25 assessment periods. The class memory unit 124 may further store overall class performance trends for instructor-level analysis. In some cases, the class memory unit 124 may be implemented using aggregation tables that are updated incrementally as individual student analyses are completed. In other cases, the class memory unit 124 may utilize materialized views that are recomputed periodically
30 based on changes to underlying learner memory units 122.

[0076] The dual-tier memory architecture 120 enables memory isolation between individual students through the use of separate learner memory units 122 for each student. The separation of the learner memory unit 122 from the class memory unit 124 permits personalized analysis without cross-contamination of student records while enabling aggregated analysis across the student cohort.

[0077] FIG. 3 illustrates an interaction between the specialized agents 130 and the working memory component 122c according to various embodiments.

[0078] As described above, the student performance analysis system 100 comprises a plurality of specialized agents 130 comprising the goal agent 132, the diagnosis agent 134, the path planner agent 136, the feedback agent 138, and an analytics agent 140. Each specialized agent may be configured to perform a distinct analytical function and write outputs to the working memory component 122c. The specialized agents 130 operate over data stored in the learner memory unit 122 and generate structured outputs that are persisted in the working memory component 122c for downstream processing.

[0079] The goal agent 132 may be configured to define measurable academic objectives based on student performance data and classification. The goal agent 132 may retrieve performance metrics from the semantic memory component 122b and historical academic events from the episodic memory component 122a to formulate personalized learning objectives. In some cases, the goal agent 132 generates primary goals and sub-goals that are written to the working memory component 122c. The primary goals may specify target performance levels for specific assessments or lab groups. The sub-goals may specify incremental milestones for achieving the primary goals. In some cases, the goal agent 132 may generate goals based on performance band classification, wherein students classified in lower performance bands receive goals focused on foundational skill improvement, and students classified in higher performance bands receive goals focused on advanced topic mastery. In other cases, the goal agent 132 may generate goals based on identified weak areas stored in the semantic memory component 122b.

[0080] The diagnosis agent 134 may be configured to identify knowledge gaps and conceptual weaknesses. The diagnosis agent 134 may perform root-cause

analysis based on data retrieved from the episodic memory component 122a and the semantic memory component 122b. In some cases, the diagnosis agent 134 identifies behavioral inefficiencies based on submission timing patterns and difficulty-level interactions stored in the episodic memory component 122a. The
5 diagnosis agent 134 may generate diagnostic insights comprising a gap severity map that indicates the severity of identified knowledge gaps across different topics or assessment areas. The diagnostic insights generated by the diagnosis agent 134 may be written to the working memory component 122c. In some cases, the diagnosis agent 134 may utilize statistical analysis techniques to identify patterns
10 in student performance data. In other cases, the diagnosis agent 134 may employ rule-based reasoning to classify knowledge gaps based on predefined criteria.

[0081] The path planner agent 136 may be configured to generate structured improvement plans comprising topic prioritization and study schedules. The path planner agent 136 may retrieve diagnostic insights from the working memory
15 component 122c and performance metrics from the semantic memory component 122b to formulate improvement plans. In some cases, the path planner agent 136 generates improvement plans comprising time allocation strategies that specify recommended study durations for different topics based on gap severity. The improvement plans may include weekly study schedules that organize learning
20 activities across multiple time periods. In some cases, the path planner agent 136 generates improvement plans comprising practice recommendations that specify types of exercises, resources, and platforms for skill development. The structured improvement plans generated by the path planner agent 136 may be written to the working memory component 122c. In some cases, the path planner agent 136 may
25 prioritize topics based on gap severity scores generated by the diagnosis agent 134. In other cases, the path planner agent 136 may prioritize topics based on upcoming assessment schedules or curriculum requirements.

[0082] The feedback agent 138 may comprise a large language model configured to generate personalized feedback messages based on data retrieved
30 from the learner memory unit 122. In some cases, the large language model may be implemented through a machine learning ecosystem using a locally deployed

language model. The large language model may comprise a Qwen2.5-Coder-7B-Instruct model deployed locally through the Hugging Face ecosystem. The feedback agent 138 may retrieve goals from the goal agent 132, diagnostic insights from the diagnosis agent 134, and improvement plans from the path planner agent 136 from the working memory component 122c to generate contextually relevant feedback. The feedback agent 138 may be configured to generate motivational guidance as part of the personalized feedback output. In some cases, the feedback agent 138 generates personalized feedback messages comprising performance summaries, identification of strengths and weaknesses, and actionable recommendations. The personalized feedback messages may range from 220 to 280 words in length. The feedback agent 138 may generate feedback that addresses the student directly and references specific performance metrics stored in the learner memory unit 122. The personalized feedback generated by the feedback agent 138 may be written to the working memory component 122c. In some cases, the feedback agent 138 may generate band-specific feedback wherein the tone and content of the feedback varies based on the student's performance band classification. In other cases, the feedback agent 138 may generate feedback that incorporates call-to-action recommendations for immediate next steps.

[0083] The analytics agent 140 may be configured to perform class-level analysis comprising performance distribution analysis, identification of weak areas across the plurality of students, and generation of instructor-level recommendations. The analytics agent 140 may retrieve data from the class memory unit 124 and aggregated outputs from the working memory components 122c of multiple students. In some cases, the analytics agent 140 may be configured to perform risk clustering analysis at the class level. The risk clustering analysis may group students based on computed risk scores and performance band classifications to identify cohorts requiring similar intervention strategies. The analytics agent 140 may generate instructor-level recommendations comprising weak student identification, class trends, intervention priorities, and lab performance bottlenecks. In some cases, the analytics agent 140 may defer execution until individual student analyses are completed, enabling the analytics

agent 140 to operate on complete individual results. In other cases, the analytics agent 140 may perform incremental analysis as individual student analyses are completed.

[0084] Each specialized agent of the plurality of specialized agents 130 may write outputs to the working memory component 122c in a structured data format. The structured data format may enable downstream processing by the output generation module 170 and retrieval by subsequent agents in the execution sequence. In some cases, the working memory component 122c may store outputs from each specialized agent in separate data fields or namespaces to maintain organization and enable selective retrieval. In other cases, the working memory component 122c may store outputs in a consolidated snapshot format that aggregates goals, diagnosis, plan, and feedback information.

[0085] FIG. 4 illustrates the educational orchestrator 150 and the agent communication layer 160 according to various embodiments.

[0086] As described above, the student performance analysis system 100 comprises the educational orchestrator 150 configured to execute the plurality of specialized agents 130 in a predetermined pedagogical sequence. The educational orchestrator 150 may coordinate multi-agent processing and manage controlled access to memory structures within the dual-tier memory architecture 120.

[0087] The educational orchestrator 150 may be configured to execute the goal agent 132, the diagnosis agent 134, the path planner agent 136, and the feedback agent 138 sequentially for each student. The sequential execution may follow a defined order wherein the goal agent 132 executes first to define measurable academic objectives, followed by the diagnosis agent 134 to identify knowledge gaps and conceptual weaknesses, followed by the path planner agent 136 to generate structured improvement plans, and followed by the feedback agent 138 to generate personalized feedback messages. The educational orchestrator 150 may be configured to execute the analytics agent 140 after completion of individual student analyses. The execution of the analytics agent 140 after completion of individual student analyses enables the analytics agent 140 to operate on complete individual results stored in the working memory components 122c of all processed students.

[0088] The educational orchestrator 150 may support scalable batch processing for processing multiple students efficiently. In some cases, the educational orchestrator 150 processes students in a batch mode wherein the sequential agent execution is performed for each student in a student cohort before the analytics agent 140 is triggered. In other cases, the educational orchestrator 150 may process students in parallel batches wherein multiple instances of the sequential agent pipeline execute concurrently for different students. The educational orchestrator 150 may implement configurable batch sizes to balance processing throughput with available computational resources.

10 [0089] In some cases, the educational orchestrator 150 may implement a priority-based execution pattern wherein students classified in lower performance bands are processed before students classified in higher performance bands. In other cases, the educational orchestrator 150 may implement a round-robin execution pattern wherein students are processed in a predetermined order regardless of performance classification. The educational orchestrator 150 may support event-driven orchestration wherein agent execution is triggered by data availability events from the data processing module 110.

15 [0090] As described above, the student performance analysis system 100 comprises the agent communication layer 160 configured to mediate inter-agent communication and maintain audit logs of agent interactions. The agent communication layer 160 may provide structured, traceable interaction between the specialized agents 130.

20 [0091] The agent communication layer 160 may be configured to provide typed message exchange between the plurality of specialized agents 130. The typed message exchange may be implemented via tuple-indexed routing. Tuple-indexed routing may comprise indexing messages using tuples that identify a sender agent, a receiver agent, and a message type. The typed message routing via tuple indexing may eliminate implicit coupling of direct agent invocation, enabling modular agent composition and replacement without modification to other agents in the system.

25 [0092] In some cases, the tuple index may further comprise a student identifier to enable

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routing of messages associated with specific students. In other cases, the tuple index may comprise a timestamp component to enable temporal ordering of messages.

[0092] The agent communication layer 160 may be configured to provide queue-based communication handling. Queue-based communication handling may enable asynchronous message delivery between the specialized agents 130. In some cases, the agent communication layer 160 implements separate message queues for each specialized agent, wherein each agent retrieves messages from a dedicated input queue. In other cases, the agent communication layer 160 implements a centralized message queue with filtering capabilities that route messages to appropriate agents based on tuple indices.

[0093] The agent communication layer 160 may be configured to provide timestamped communication records. The timestamped communication records may comprise a timestamp indicating when each message was sent, a timestamp indicating when each message was received, and a timestamp indicating when each message was processed. The timestamped communication records may enable chronological reconstruction of agent interactions for debugging and verification purposes.

[0094] The agent communication layer 160 may implement append-only provenance logging that produces a causally-ordered audit trail satisfying tamper-evidence requirements. The append-only provenance logging may record each message exchange in a log structure that does not permit modification or deletion of previously recorded entries. The causally-ordered audit trail may preserve the causal relationships between messages, enabling reconstruction of the reasoning sequence performed by the specialized agents 130. In some cases, the append-only provenance logging may implement cryptographic hashing to detect unauthorized modifications to the audit trail. The tamper-evidence requirements may support academic integrity by providing verifiable records of analytical decisions made by the system.

[0095] The agent communication layer 160 may implement pull-semantics cohort aggregation whereby the analytics agent 140 defers message consumption until the entire student cohort has been processed. Pull-semantics cohort

aggregation may enable the analytics agent 140 to retrieve aggregated outputs from the working memory components 122c of all students in a cohort before performing class-level analysis. In some cases, the agent communication layer 160 maintains a cohort completion indicator that signals when all students in a cohort have been processed by the individual student-focused agents. The analytics agent 140 may poll the cohort completion indicator before initiating message consumption. In other cases, the agent communication layer 160 may implement a callback mechanism that notifies the analytics agent 140 when cohort processing is complete.

10 **[0096]** In some cases, the agent communication layer 160 may implement message prioritization wherein messages associated with at-risk students are processed before messages associated with other students. In other cases, the agent communication layer 160 may implement message batching wherein multiple messages are aggregated and delivered together to reduce communication overhead. The agent communication layer 160 may support configurable retention policies for audit logs, enabling storage management while preserving records for compliance purposes.

15 **[0097]** FIG. 5 illustrates a method 500 for student performance analysis according to various embodiments.

20 **[0098]** The method 500 may be performed by the student performance analysis system 100 described above. The method 500 comprises a sequence of steps that transform structured academic data into personalized reports and class-level insights through coordinated agent execution.

25 **[0099]** At a step 502, the method 500 comprises receiving structured academic data associated with a plurality of students. The structured academic data may comprise student identifiers, assessment details, marks obtained, submission duration data, lab categories, and difficulty levels. In some cases, the step 502 comprises processing the structured academic data through a data processing pipeline to generate enriched student profile data. The data processing pipeline may perform extraction and normalization of lab identifiers and difficulty levels,

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computation of percentage scores, detection of rapid or abnormal submissions, and mapping of labs into predefined groups.

5 **[0100]** At a step 504, the method 500 comprises building learner memory units for each student of the plurality of students. The step 504 may comprise populating an episodic memory component with historical academic events, populating a semantic memory component with derived performance metrics, and initializing a working memory component for storing analysis outputs. In some cases, the step 504 comprises computing performance band classifications and risk scores for each student based on the enriched student profile data.

10 **[0101]** At a step 506, the method 500 comprises building a class memory unit based on aggregated data from the plurality of students. The step 506 may comprise computing class-level statistics including total number of students, distribution across performance bands, and group-level averages. In some cases, the step 506 comprises identifying at-risk students and detecting abnormal submission patterns across the plurality of students.

15 **[0102]** At a step 508, the method 500 comprises executing a plurality of specialized agents in a predetermined pedagogical sequence for each student. The step 508 may comprise executing a goal agent to define measurable academic objectives, executing a diagnosis agent to identify knowledge gaps and conceptual weaknesses, executing a path planner agent to generate structured improvement plans, and executing a feedback agent to generate personalized feedback messages. In some cases, each specialized agent writes outputs to the working memory component of the corresponding learner memory unit. The agent communication layer 160 may log each inter-agent communication during the step 508.

25 **[0103]** At a step 510, the method 500 comprises executing an analytics agent to perform class-level analysis. The step 510 may be performed after completion of individual student analyses at the step 508. The analytics agent may retrieve data from the class memory unit and aggregated outputs from the working memory components of multiple students. In some cases, the step 510 comprises generating instructor-level recommendations comprising weak student identification, class trends, intervention priorities, and performance bottlenecks.

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[0104] At a step 512, the method 500 comprises generating personalized reports based on outputs stored in the working memory components. The step 512 may comprise generating at least one of a portable document format report, a spreadsheet report, and an electronic mail notification. In some cases, the step 512
5 comprises generating band-specific personalized notifications containing student performance summaries, study roadmaps, and call-to-action recommendations.

[0105] In some cases, the method 500 may be performed in alternative sequences. The step 504 and the step 506 may be performed in parallel, wherein learner memory units and the class memory unit are built concurrently. In other
10 cases, the step 506 may be performed incrementally during the step 504, wherein class-level statistics are updated as each learner memory unit is built.

[0106] In some cases, the step 508 may be performed using parallel processing for multiple students. Multiple instances of the sequential agent pipeline may execute concurrently for different students. In other cases, the step 508 may be
15 performed using batch processing wherein students are grouped into batches and each batch is processed sequentially. The method 500 may implement priority-based execution wherein students classified in lower performance bands are processed before students classified in higher performance bands.

[0107] In some cases, the step 510 may be performed incrementally during the
20 step 508, wherein the analytics agent performs partial class-level analysis as individual student analyses are completed. In other cases, the step 510 may be deferred until all students in a cohort have been processed at the step 508.

[0108] In some cases, the step 512 may be performed concurrently with the
25 step 510, wherein personalized reports for individual students are generated while class-level analysis is being performed. In other cases, the step 512 may be performed after completion of both the step 508 and the step 510, enabling the personalized reports to incorporate class-level insights.

[0109] FIG. 6 illustrates the agent communication layer 160 according to various embodiments.

30 [0110] The agent communication layer 160 comprises an internal structure configured to facilitate typed, audit-logged communication between the specialized

agents 130. The agent communication layer 160 may comprise a message routing component, a message queue component, an audit logging component, and a cohort aggregation component.

[0111] The message routing component may be configured to route messages
5 between the specialized agents 130 based on tuple indices. Each tuple index may comprise a sender agent identifier, a receiver agent identifier, and a message type identifier. In some cases, the tuple index may further comprise a student identifier to enable routing of messages associated with specific students. In other cases, the tuple index may comprise a timestamp component to enable temporal ordering of
10 messages. The message routing component may maintain a routing table that maps tuple indices to destination agents.

[0112] The message queue component may be configured to store messages pending delivery to the specialized agents 130. In some cases, the message queue component implements separate message queues for each specialized agent,
15 wherein each agent retrieves messages from a dedicated input queue. In other cases, the message queue component implements a centralized message queue with filtering capabilities that route messages to appropriate agents based on tuple indices. The message queue component may implement first-in-first-out ordering for messages within each queue. In some cases, the message queue component may
20 implement priority-based ordering wherein messages associated with at-risk students are positioned ahead of messages associated with other students.

[0113] The audit logging component may be configured to record each message exchange in an append-only log structure. The append-only log structure may not permit modification or deletion of previously recorded entries. Each log
25 entry may comprise a timestamp indicating when the message was sent, a timestamp indicating when the message was received, a timestamp indicating when the message was processed, the tuple index of the message, and the message content. In some cases, the audit logging component may implement cryptographic hashing to detect unauthorized modifications to the audit trail. The cryptographic
30 hashing may compute a hash value for each log entry based on the log entry content

and the hash value of the preceding log entry, creating a hash chain that enables detection of tampering.

[0114] The cohort aggregation component may be configured to support pull-semantics cohort aggregation for the analytics agent 140. The cohort aggregation component may maintain a cohort completion indicator that tracks the number of students processed by the individual student-focused agents. In some cases, the cohort aggregation component signals the analytics agent 140 when all students in a cohort have been processed. In other cases, the cohort aggregation component may implement a callback mechanism that notifies the analytics agent 140 when cohort processing is complete.

[0115] In some cases, the agent communication layer 160 may implement message batching wherein multiple messages are aggregated and delivered together to reduce communication overhead. The message batching may aggregate messages destined for the same receiver agent within a configurable time window. In other cases, the agent communication layer 160 may implement message compression to reduce storage requirements for the audit log.

[0116] The agent communication layer 160 may support configurable retention policies for audit logs. The retention policies may specify a retention period after which audit log entries may be archived or deleted. In some cases, the retention policies may specify different retention periods for different message types or different performance bands. The agent communication layer 160 may implement log rotation wherein older audit log entries are moved to archival storage while recent entries remain in active storage.

[0117] In some cases, the agent communication layer 160 may implement message validation wherein incoming messages are validated against predefined schemas before being routed to destination agents. The message validation may verify that message content conforms to expected data structures for the specified message type. In other cases, the agent communication layer 160 may implement message transformation wherein messages are transformed from one format to another during routing.

[0118] It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described herein, but rather may be useful in substantially any application where it is desired to analyze performance data and generate personalized feedback through coordinated multi-agent processing. The student performance analysis system 100 and associated methods are susceptible of industrial application in the education technology industry, including deployment within universities, colleges, schools, vocational training institutions, and online learning platforms. The system may be implemented within learning management systems, academic information systems, and institutional analytics platforms to provide automated performance analysis and intervention capabilities.

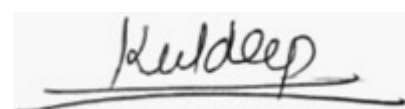
[0119] While particular embodiments and applications of the present disclosure have been illustrated and described in connection with student academic performance analysis and the education industry, it is to be understood that embodiments of the disclosure are not limited to the precise construction and compositions disclosed herein. The multi-agent architecture, dual-tier memory structure, and orchestrated agent execution patterns described herein may find application in other domains where personalized analysis and feedback generation are desired, such as employee training and development systems, professional certification programs, and skill assessment platforms. Various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims.

[0120] There are several technical advantages of the student performance analysis system of the present disclosure. First, the dual-tier memory architecture separating learner-specific memory from class-level memory enables memory isolation between individual students while permitting aggregated analysis across the student cohort, thereby allowing personalized analysis without cross-contamination of student records. Second, the decomposition of analytical functions into specialized agents, each performing a distinct task such as goal-setting, diagnosis, planning, and feedback generation, enables modular reasoning

where each analytical task produces structured intermediary artifacts that can be inspected, audited, or used by subsequent processing components. Third, the predetermined pedagogical sequence enforced by the educational orchestrator ensures that agents operate in a logical order that mirrors effective educational intervention workflows, enabling downstream agents to build upon the outputs of upstream agents. Fourth, the agent communication layer with typed message exchange, queue-based communication handling, and timestamped communication records provides traceability and explainability for every analytical decision made by the system, supporting academic integrity and enabling post-hoc verification of the reasoning process. Fifth, the use of a large language model within the feedback agent enables context-aware reasoning and natural language generation, allowing the system to produce personalized feedback messages that are tailored to each student's specific academic profile and learning history rather than relying on generic template-based responses. Sixth, the append-only provenance logging with cryptographic hashing produces a causally-ordered audit trail satisfying tamper-evidence requirements, enabling detection of unauthorized modifications to analytical records. Additionally, the support for local deployment enables the system to operate on institutional computing resources without external network dependencies, addressing data privacy considerations in educational environments. These technical advantages collectively contribute to a scalable and intelligent academic intervention framework that provides both individualized student-level analysis and aggregated class-level insights for instructors.

[0121] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This day of May 2026

A handwritten signature in black ink, appearing to read "Kuldeep", is written over a horizontal line.

(Kuldeep Singh)

Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358

CLAIMS

WE CLAIM:

1. A student performance analysis system (100), comprising:
 - a data processing module (110) configured to receive structured academic data associated with a plurality of students and generate enriched student profile data;
 - a dual-tier memory architecture (120) comprising:
 - a learner memory unit (122) associated with each student of the plurality of students, wherein the learner memory unit (122) comprises an episodic memory component (122a) configured to store historical academic events, a semantic memory component (122b) configured to store derived performance metrics, and a working memory component (122c) configured to store intermediate and final analysis outputs; and
 - a class memory unit (124) configured to store aggregated class-level data;
 - a plurality of specialized agents (130) comprising a goal agent (132), a diagnosis agent (134), a path planner agent (136), a feedback agent (138), and an analytics agent (140), wherein each specialized agent is configured to perform a distinct analytical function and write outputs to the working memory component (122c);
 - an educational orchestrator (150) configured to execute the plurality of specialized agents (130) in a predetermined pedagogical sequence;
 - an agent communication layer (160) configured to mediate inter-agent communication and maintain audit logs of agent interactions; and
 - an output generation module (170) configured to generate personalized reports based on the outputs stored in the working memory component (122c).
2. The system (100) of claim 1, wherein the feedback agent (138) comprises a large language model configured to generate personalized feedback messages based on data retrieved from the learner memory unit (122).
3. The system (100) of claim 1, wherein the agent communication layer (160) is configured to provide typed message exchange between the plurality of

specialized agents (130), queue-based communication handling, and timestamped communication records.

4. The system (100) of claim 1, wherein the goal agent (132) is configured to define measurable academic objectives, the diagnosis agent (134) is
5 configured to identify knowledge gaps and conceptual weaknesses, and the path planner agent (136) is configured to generate structured improvement plans comprising topic prioritization and study schedules.

5. The system (100) of claim 1, wherein the analytics agent (140) is configured to perform class-level analysis comprising performance distribution
10 analysis, identification of weak areas across the plurality of students, and generation of instructor-level recommendations.

6. The system (100) of claim 1, wherein the episodic memory component (122a) is configured to store assessment attempts, scores obtained, submission time data, and difficulty-level interactions for each student.

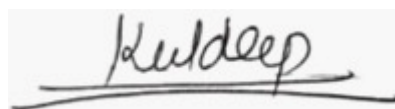
15 7. The system (100) of claim 1, wherein the semantic memory component (122b) is configured to store a performance band classification, group-wise proficiency data, and a computed risk score for each student.

8. The system (100) of claim 1, further comprising a performance classification module (180) configured to classify each student into one of a
20 plurality of performance bands based on an overall performance score.

9. The system (100) of claim 1, wherein the output generation module (170) is configured to generate at least one of a portable document format report, a spreadsheet report, and an electronic mail notification comprising a student performance summary and a study roadmap.

25 10. The system (100) of claim 1, wherein the educational orchestrator (150) is configured to execute the goal agent (132), the diagnosis agent (134), the path planner agent (136), and the feedback agent (138) sequentially for each student, and to execute the analytics agent (140) after completion of individual student analyses.

30 **Dated This day of May 2026**

A handwritten signature in black ink, reading "Kuldeep", is written over a horizontal line. The signature is enclosed in a light gray rectangular box.

(Kuldeep Singh)

Authorized Agent for the Applicant,

Indian Patent Agent Regn No. IN/PA-4358

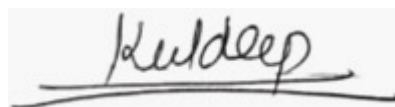
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ABSTRACT

MULTI-AGENT SYSTEM FOR STUDENT PERFORMANCE ANALYSIS AND AUTOMATED INTERVENTION

A student performance analysis system (100) comprises a data processing module
5 (110) configured to receive structured academic data associated with a plurality of
students and generate enriched student profile data, and a dual-tier memory
architecture (120) comprising a learner memory unit (122) having an episodic
memory component (122a), a semantic memory component (122b), and a working
memory component (122c), and a class memory unit (124). The system (100)
10 comprises specialized agents (130) including a goal agent (132), diagnosis agent
(134), path planner agent (136), feedback agent (138), and analytics agent (140),
each writing outputs to the working memory component (122c). An educational
orchestrator (150) executes the specialized agents (130) in a predetermined
pedagogical sequence. An agent communication layer (160) mediates inter-agent
15 communication and maintains audit logs. An output generation module (170)
generates personalized reports based on outputs in the working memory component
(122c).

Dated This day of May 2026



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